

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • GPA: 3.75/4.0

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

WORK EXPERIENCE

Google | Software Engineer Intern — Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers serving **3B+ active users** at **sub-50ms p99, 10K+ req/sec** — selected Protobuf over custom serialization for schema evolution and cross-language compatibility
- Profiled settings navigation as p99 bottleneck at 1,200ms; implemented **lock-free concurrent trie search** eliminating mutex contention — **96% latency reduction**, zero production regressions in Chrome stable channel
- Architected event-driven **TypeScript/React** system with observer pattern across **25K+ lines** of Chromium at **95% test coverage**; design documents adopted into production branch by senior engineers

CoderPush | Software Engineer Intern

Sep 2023 – Dec 2023

- Diagnosed DynamoDB hot-partition failure at **9,000+ req/sec**; redesigned partition key strategy and access patterns — **30% read throughput improvement** under sustained high-concurrency traffic
- Designed idempotent payment APIs with Redis distributed caching (**85% cache hit rate**) and exponential backoff — enforced exactly-once semantics preventing duplicate transactions under network partitions

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on AWS — selected JWT over session auth for horizontal scalability; auto-scaling supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+; redesigned **PostgreSQL** indexing and query structure achieving **sub-100ms** response for 10,000+ records

PERSONAL PROJECTS

TiMoto AI – Backend Platform & Distributed Systems | *Go, gRPC, Python, AWS, K8s, Terraform*timoto.ai

- **Sole engineer** — designed, built, and operate end-to-end backend platform: **gRPC** inter-service layer, Python/Django REST API, PostgreSQL, and multi-AZ **AWS** infrastructure
- Resolved **production deadlock under concurrent gRPC calls** — traced root cause to circular resource acquisition across services; redesigned call sequencing — **100% success rate** at **sub-50ms p99**
- Architected multi-AZ ECS Fargate with Terraform IaC, circuit breaker pattern, and health-check-triggered auto-rollback — **99.9% uptime**, **44% cost reduction** to \$40–60/month
- Integrated vLLM inference engine; diagnosed and eliminated KV cache memory fragmentation under concurrent load via **PagedAttention** — zero OOM failures at production traffic

Pulumi – Open Source Contributor | *Go, TypeScript, IaC*

24.4K★ github.com/pulumi/pulumi

- Submitted **Go** CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning — under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer — studied linearizability guarantees and correctness under concurrent operations and partial failures

TECHNICAL SKILLS

Languages: C++, Go, Python, TypeScript, Java, JavaScript, SQL, Bash, Rust

Distributed Systems: gRPC, Protocol Buffers, Circuit breakers, Exactly-once semantics, Raft/Paxos, Fault tolerance, Event-driven architecture

Cloud & Infrastructure: AWS (ECS Fargate, EC2, S3, RDS), Kubernetes, Docker, Terraform, CI/CD, GitHub Actions, GCP

Frameworks & Databases: Django, FastAPI, Node.js, React, REST APIs, PostgreSQL, MySQL, Redis, DynamoDB, MongoDB

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LLM-as-a-judge evaluation, PyTorch, Ray